

SHAHADAT HUSSAIN, PhD

MATERIALS & MANUFACTURING ENGINEER — ADDITIVE MANUFACTURING / METAMATERIALS

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PROFESSIONAL SUMMARY

Results-driven Materials Scientist and Manufacturing Engineer with 6+ years of experience bridging advanced research and industrial application in the UAE. Expertise spans the full product development lifecycle of additive manufacturing (LPBF, FFF, SLA), from experimental design and materials selection to prototype refinement and manufacturability validation. Hands-on proficiency with industrial 3D printing systems and metallurgical characterization (SEM, XRD, DSC, Instron, MTS), with a specialized focus on TPMS metamaterials and shape memory alloys. Combines deep materials engineering knowledge with Python/MATLAB data analysis to drive root-cause investigation, defect reduction, and process optimization. Seeking to leverage deep technical expertise in experimental design, prototyping, and scale-up to drive innovation in advanced R&D or manufacturing engineering roles.

CORE COMPETENCIES

- Additive Manufacturing Process Development (LPBF, FFF, SLA)
- Mechanical & Thermal Qualification Testing (Static, Fatigue, Compression, Vibration/Modal)
- Materials Characterization (SEM, XRD, DSC, Optical Microscopy)
- Architected Lattice Structures for Thermal & Structural Performance (TPMS)
- Process Parameter Optimization & Defect/Quality Mapping
- NiTi & Cu-Based Shape Memory Alloys
- CAD & Build Preparation (SolidWorks, FreeCAD, Materialise Magics, GrabCAD)
- Data Analysis & Process Modeling (Python, MATLAB, SQL)
- EHS / Lab Safety & Risk Assessment

PROFESSIONAL EXPERIENCE

Associate Research Scientist — Additive Manufacturing & Advanced Materials

Sep 2024 – Present

New York University Abu Dhabi (NYUAD), Abu Dhabi, UAE

- Delivered carbon-fiber-reinforced PLA/PPA composite lattice components (gyroid and primitive TPMS geometries) via FFF, engineered for lightweight, high-strength, vibration-damping performance relevant to structural and thermal hardware applications.
- Qualified printed composite structures against quasi-static compression, flexural, modal, and frequency-response test protocols to validate strength, energy absorption, and vibration performance under service-like loading.
- Operated and maintained 10+ industrial additive manufacturing systems (Raise3D Pro3 Plus HS, Stratasys J850/F370, Formlabs Form3, Bambu Lab Carbon X1), sustaining continuous production and prototyping throughput.

Postdoctoral Research Fellow — Metal 3D Printing & Smart Materials

Feb 2020 – Jun 2024

Khalifa University, Abu Dhabi, UAE

- Fabricated and qualified NiTi shape memory alloy TPMS lattice structures via Laser Powder Bed Fusion (EOS M400); optimized laser power, scan speed, and energy density to control microstructure, porosity, and phase transformation behavior — a process-control approach directly transferable to defect reduction in production manufacturing.
- Executed a full materials characterization pipeline on 50+ printed samples using SEM, FIB-SEM (SEI/BSE/EDS), XRD, DSC, and optical microscopy to link process parameters to structural and functional performance.
- Ran static (Instron 5969, 50 kN), fatigue (Instron 8872, 25 kN), and compression (MTS 100 kN) qualification campaigns; developed defect and print-quality maps used to reduce porosity and improve part consistency — a qualification methodology applicable to reliability testing of power and thermal hardware.
- Authored 4 peer-reviewed publications; completed comprehensive EHS lab safety training covering chemical handling, cryogenic substances, risk assessment, and emergency response protocols.

Project Fellow — Smart & Functional Materials

Jun 2013 – Aug 2014

CSIR – Advanced Materials & Processes Research Institute (AMPRI), Bhopal, India

- Synthesized Cu-Al-Mn and Cu-Al-Ni shape memory alloys via vacuum induction melting under a government-funded national research program; performed hot/cold rolling, heat treatment, quenching, and metallographic preparation.
- Characterized phase transformation and mechanical properties using SEM, XRD, DSC, and hardness/tensile testing; contributed to 5 peer-reviewed publications and a research collaboration with IIT Madras.

Engineering Intern — Steel Plant Operations

Jun – Jul 2011

Bokaro Steel Plant, Steel Authority of India Limited (SAIL), Jharkhand, India

- Gained direct industrial exposure to large-scale integrated steel manufacturing processes, plant operations, and quality systems.

EDUCATION

PhD, Materials Science & Technology | 2014 – 2019

Academy of Scientific and Innovative Research (AcSIR) / CSIR-AMPRI, Bhopal, India — Thesis: Cu-based shape memory alloys: composition, processing, microstructure, and shape memory properties

BEng, Mechanical Engineering | 2008 – 2012

UIT, Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal, India

TECHNICAL SKILLS

Additive Manufacturing Systems: EOS M400 (LPBF/SLM); Stratasys F370/J850 (FDM/PolyJet); Raise3D Pro3 Plus HS, Forge1 (FFF); Formlabs Form3 (SLA); Ultimaker S1

Materials Characterization: JEOL 7610F SEM; FEI Nova NanoSEM; FEI Quanta 3D FIB-SEM (SEI/BSE/EDS); Bruker D2 & Rigaku MiniFlex XRD; Mettler Toledo & Setaram DSC; Leica Optical Microscopy

Mechanical Testing: Instron 5969 (static, 50 kN); Instron 8872 (fatigue, 25 kN); MTS 100 kN UTM; Controls Simplex 350 & Automax

Design & Computing: SolidWorks, FreeCAD, Materialise Magics, GrabCAD Print; Python (Pandas, NumPy, Scikit-learn), MATLAB, SQL, Linux

CERTIFICATIONS

- Additive Manufacturing Specialization — Arizona State University
- IBM Data Science Professional Certificate — IBM
- Google AI Essentials Certificate — Google
- Lab Safety (EHS) — Khalifa University

PUBLICATIONS & RECOGNITION

15+ peer-reviewed publications | 500+ citations | h-index: 10 | Peer reviewer for 15+ journals including Additive Manufacturing, Materials & Design, Scripta Materialia. Full list: scholar.google.com/citations?user=tQNSWaAAAAAJ

PROFESSIONAL AFFILIATIONS

Member, AM-Platform: European Technology Platform in Additive Manufacturing | Associate Member, European Materials Modelling Council | Individual Member, Research Data Alliance | Technical Volunteer, Materials Research Society

ADDITIONAL INFORMATION

Languages: English (Professional), Hindi (Native), Urdu (Native) | Profiles: github.com/drshahadathussain | researchgate.net/profile/Shahadat-Hussain